Is Matter Around Us Pure?

The big idea

In everyday language "pure" means unadulterated. To a scientist, a **pure substance** has particles that are all the same chemical: an **element** or a **compound**. Most things around us are **mixtures**.

Mixtures

- **Homogeneous** (uniform composition): salt in water, air, alloys. These are **solutions**.
- **Heterogeneous** (non-uniform): sand in water, oil and water, soil.

Type	Particle size	Visible?	Settles?	Tyndall effect?	Example
Solution	Smallest (< 1 nm)	No	No	No	Sugar in water
Colloid	Intermediate	No (to the eye)	No	**Yes**	Milk, fog, smoke, jelly
Suspension	Large	Yes	Yes	Yes	Chalk in water, muddy water

Concentration of a solution = amount of solute / amount of solution (mass by mass % or mass by volume %). A **saturated** solution can't dissolve any more solute at that temperature; **solubility** is the amount dissolved in 100 g of solvent.

Separating mixtures

Method	Separates	Example
Evaporation	Dissolved solid from liquid	Salt from sea water
Centrifugation	Fine suspended particles by spinning	Cream from milk
Separating funnel	Immiscible liquids	Oil and water
Sublimation	Sublimable solid from others	Ammonium chloride from salt
Chromatography	Coloured components	Dyes in ink
Distillation	Miscible liquids with very different boiling points	Acetone and water
Fractional distillation	Liquids with close boiling points	Gases of air, petroleum
Crystallisation	Pure solid from solution	Pure copper sulphate, salt

Physical and chemical changes

- **Physical change**: no new substance forms; often reversible (melting ice, cutting paper).
- **Chemical change**: a new substance forms (burning, rusting, cooking).

Elements and compounds

- **Element**: cannot be broken into simpler substances by chemical reactions: metals, non-metals, metalloids (boron, silicon).
- **Compound**: two or more elements **chemically combined** in a **fixed ratio**. Its properties differ from its elements (iron + sulphur heated → iron sulphide, which isn't magnetic).

MIXTURE VS COMPOUND

Mixture: variable composition, components keep their properties, separated by physical methods.
Compound: fixed composition, new properties, separated only by chemical methods.

Quick recap

- Pure substances: elements and compounds. Mixtures: homogeneous or heterogeneous.
- Solution, colloid and suspension differ in particle size; colloids show the Tyndall effect.
- Choose the separation method by the property that differs.

Practice questions

1. Which of these shows the Tyndall effect?
   (a) Salt solution (b) Sugar solution (c) Milk (d) Copper sulphate solution
2. Cream is separated from milk by
   (a) filtration (b) centrifugation (c) distillation (d) sublimation
3. Which is a chemical change?
   (a) Melting of wax (b) Dissolving sugar (c) Burning of paper (d) Freezing of water
4. Which is a compound?
   (a) Air (b) Brass (c) Water (d) Sea water
5. Dyes in black ink can be separated by
   (a) filtration (b) chromatography (c) centrifugation (d) sedimentation

Answers: 1. c 2. b 3. c 4. c 5. b